



NLP Project Proposal

TempoRAG-KG

Temporal Knowledge Graph-Augmented Retrieval
for Multi-Hop Question Answering

Supanut Kompayak · Aphisit Jaemyaem · Dechathon Niamsa-ard · Kaung Hein Htet

Asian Institute of Technology

We noticed a gap in KG²RAG

(Soman et al., 2024)

What the Knowledge Graph stores:

Steve Jobs · CEO_of · Apple

Tim Cook · CEO_of · Apple

⚠ No temporal info — both look equally valid

Query:
"Who was CEO
of Apple
in 2005?"

System response:

Retrieves both facts

↓
CONFLICT

No way to decide

↓
X Hallucinates

Key observation: KG triples are stored as timeless facts. the retriever has no mechanism to resolve which is valid at query time.

Why It Matters

35.0%

HotpotQA questions
have temporal expressions

38.3%

MuSiQue questions
have temporal expressions

46.7%

in 4-hop questions
↑ harder = more temporal

The Research Gap

No existing KG-RAG system — including KG²RAG, GEAR, GraphRAG, or GoG — attaches temporal validity to KG triples **or filters stale facts by time**.

TempoRAG-KG is the first framework to do this on general-domain unstructured text.

Our Solution: TempoRAG-KG

① Temporal Triple Tagging

TempAgent [Xiong et al., NAACL 2025]

Attach [valid_from, valid_to] to every KG triple at construction time — enabling deterministic point-in-time filtering during retrieval.

② GoG Fill-in Mechanism

GoG [He et al., EMNLP 2024]

When temporal filtering leaves dead ends in the graph, invoke LLM parametric knowledge to fill in missing facts (confidence threshold ≥ 0.7).

Temporal Filter (applied before GEAR graph expansion)

Keep edge e iff: $(\text{valid_from} = \text{null} \text{ OR } \text{valid_from} \leq y_q) \text{ AND } (\text{valid_to} = \text{null} \text{ OR } \text{valid_to} \geq y_q)$

Temporal Failure Taxonomy (4 types — all addressed by TempoRAG-KG)

Type 1

Stale Fact

Type 2

Conflicting Facts

Type 3

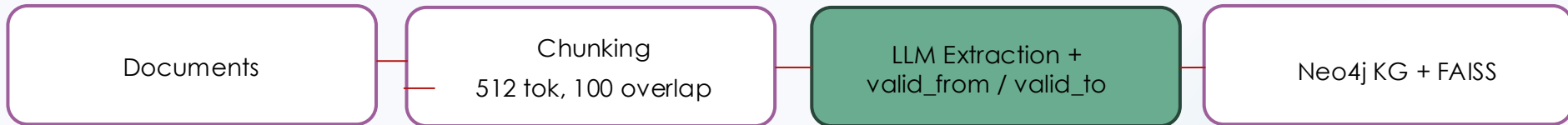
Missing Context

Type 4

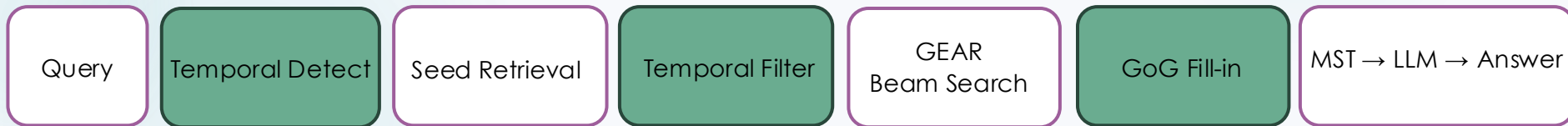
Hop Failure

System Architecture

OFFLINE



ONLINE



Our additions



KG²RAG backbone

Evaluation Plan

Datasets

HotpotQA

7,405 questions | 2-hop | 35.0% temporal

MuSiQue

2,417 questions | 2–4 hop | 38.3% temporal

Metrics

F1 & EM (overall + temporal / non-temporal subsets)

Retrieval Precision@5 (gold supporting facts retrieved)

Temporal Extraction F1 (RQ3, ± 1 year tolerance)

Ablation Design

Condition	Temporal	GoG	Graph Exp.
Vanilla RAG	✗	✗	✗
KG ² RAG	✗	✗	✓
+Temporal only	✓	✗	✓
+GoG only	✗	✓	✓
Full (ours)	✓	✓	✓

Table 2: Ablation conditions. ✓ = component enabled; ✗ = disabled.

Expected Results Table

Model	Overall		Temporal Subset		Non-temporal Subset	
	F1	EM	F1	EM	F1	EM
Vanilla RAG	–	–	–	–	–	–
GraphRAG	–	–	–	–	–	–
KG ² RAG	–	–	–	–	–	–
TempoRAG-KG (ours)	–	–	–	–	–	–

Progress & Takeaway

✓ Completed

- Literature review (16 papers)
- EDA: HotpotQA + MuSiQue
- Temporal failure taxonomy (4 types)
- Competitor analysis (EvoReasoner)

▶ In Progress

- Neo4j schema + setup
- Temporal extraction prompts
- GitHub repository setup

□ Remaining

- KG²RAG baseline reproduction
- Manual annotation (200 passages)
- Full pipeline implementation
- Ablation experiments
- Streamlit demo

"Temporal validity should be a first-class property of KG edges — not an optional annotation."